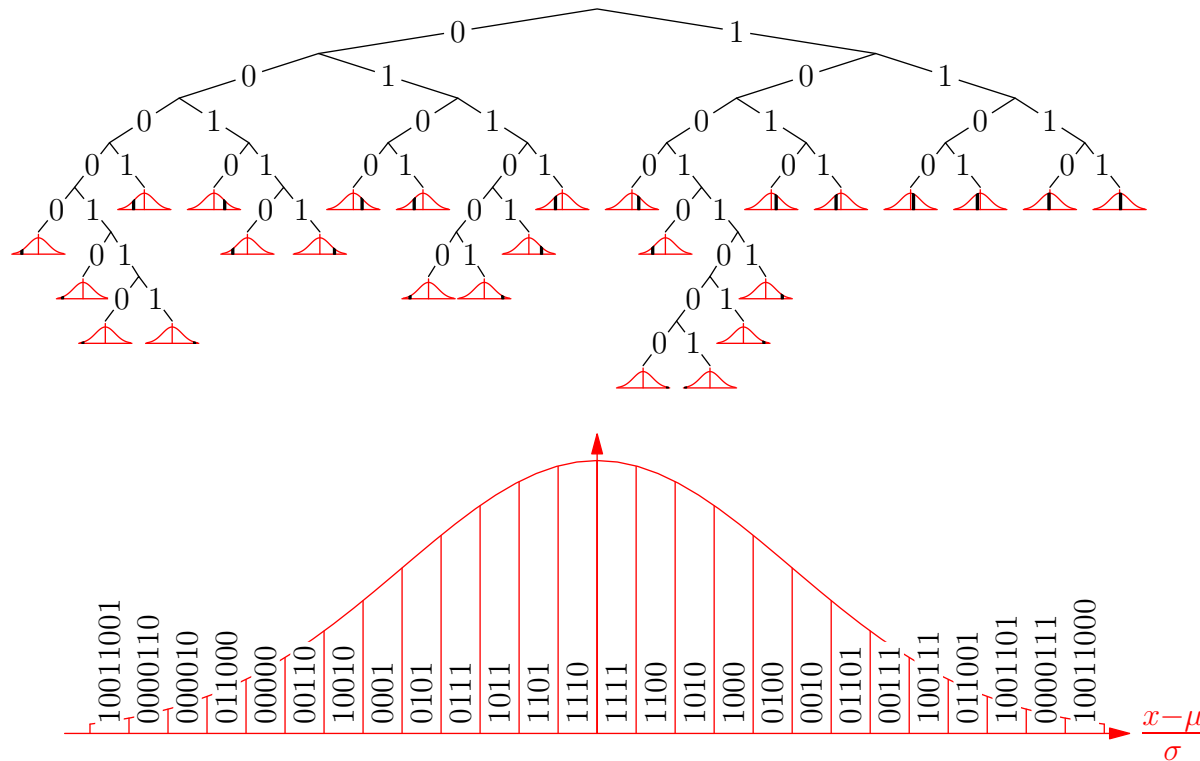


# Mathematics of Machine Learning

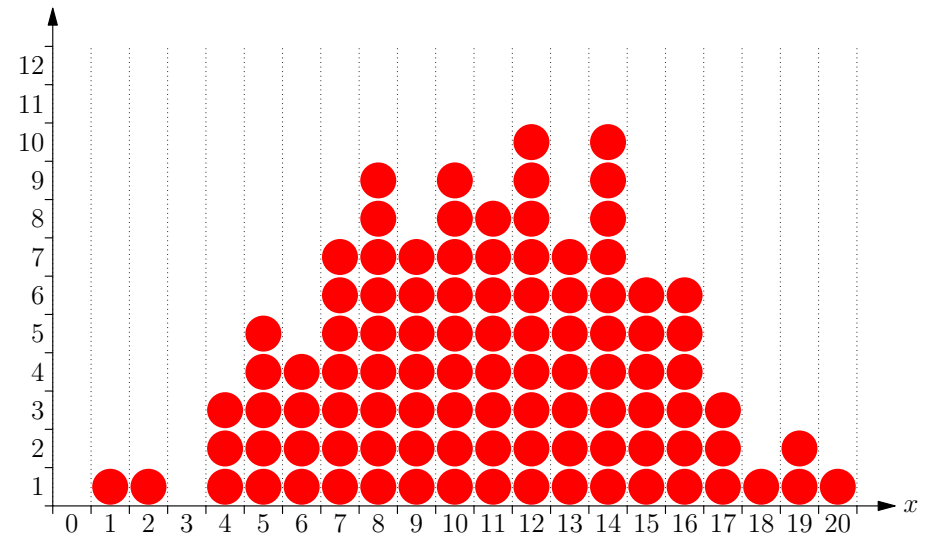
## *Entropy*



## *Entropy, Coding, Maximum Entropy*

# Outline

1. **Measuring Uncertainty**
2. Code Length
3. Maximum Entropy



# Measuring Uncertainty

- What is more uncertain tossing a coin three times or throwing a dice
- The answer depends on whether you care about the order of the coin tosses
- But, how do we answer such a question?
- Let  $X$  be a random variable denoting the possible outcomes
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$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))$$

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# Let's Calculate

- For an honest dice  $D \in \{1,2,3,4,5,6\}$  and  $\mathbb{P}(D = i) = 1/6$  so

$$H_D = -\sum_{i=1}^6 \frac{1}{6} \log_2 \left( \frac{1}{6} \right) = -\log_2 \left( \frac{1}{6} \right) = \log_2(6) \approx 2.584 \text{bits}$$

- For an honest coin where we care about the order so  $C \in \{000,001,\dots,111\}$  the  $\mathbb{P}(C = i) = \frac{1}{8}$  and

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- What if we don't care about the order of the out-come then  $\mathbb{P}(HHH) = \mathbb{P}(TTT) = 1/8$ ,  $\mathbb{P}(HHT) = \mathbb{P}(HTT) = 3/8$  so

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- But why Shannon entropy?

# Additive Entropy

- If  $H_X$  and  $H_Y$  is the uncertainty of two independent random variable  $X$  and  $Y$ , what is the uncertainty of the combined event  $(X,Y)$ ?

$$\begin{aligned} H_{(X,Y)} &= - \sum_{X,Y} \mathbb{P}(X,Y) \log_2(\mathbb{P}(X,Y)) \\ &= - \sum_{X,Y} \mathbb{P}(X)\mathbb{P}(Y) \log_2(\mathbb{P}(X)\mathbb{P}(Y)) \\ &= - \sum_{X,Y} \mathbb{P}(X)\mathbb{P}(Y) (\log_2(\mathbb{P}(X)) + \log_2(\mathbb{P}(Y))) \\ &= - \sum_X \mathbb{P}(X) \log_2(\mathbb{P}(X)) - \sum_Y \mathbb{P}(Y) \log_2(Y) = H_X + H_Y \end{aligned}$$

- Shannon's entropy is one of the few functions that satisfy this condition

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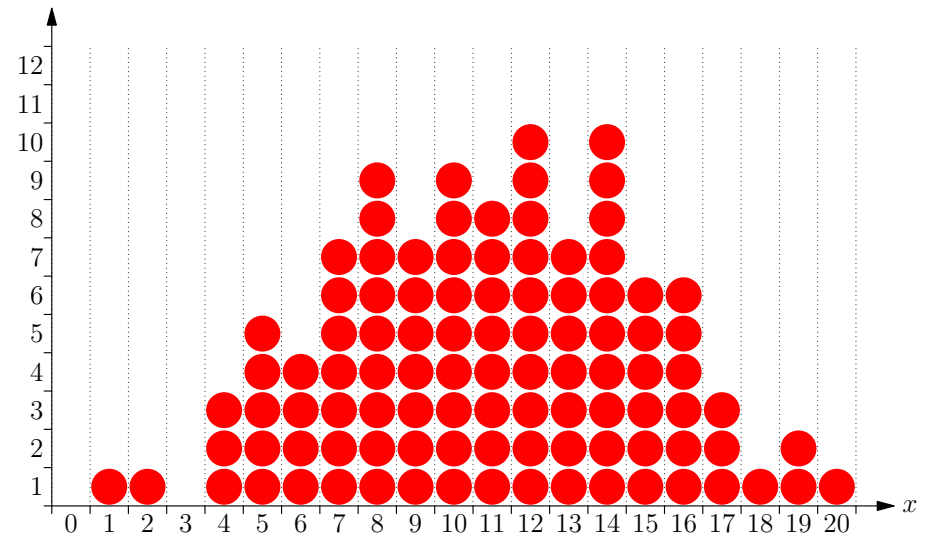
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- Suppose we had to communicate a message with  $2^n$  equally likely outputs (e.g. the result of  $n$ -coin tosses)
- We can do this with a binary string with  $n$  bits (011..0)
- If there were 5 possible outcomes I could do this with 3 bits, but waste  $3/8$  of the message
- However if we have a batch of 3 independent messages each with 5 outcomes then there are 125 possible outcomes
- By batching together enough messages with  $N$  outcomes then we asymptotically need just  $\log_2(N)$  bits

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# Different Probabilities

- We “showed” that if we had  $N$  events,  $X_i$ , each with probability,  $\mathbb{P}(X_i) = 1/N$ , we can code the outcomes with a message of length  $-\log_2(\mathbb{P}(X_i))$
- With a shorter message we would not be able to distinguish all possible outcomes from the message
- What happens if some of outcomes occur with a different probability

|                         |               |               |               |               |               |               |
|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $X_i$ :                 | 1             | 2             | 3             | 4             | 5             | 6             |
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- If the probabilities are not equal to  $2^{-n}$  we can still find a code with a length very close to  $-\log_2(\mathbb{P}(X))$  per message by transmitting a large number of messages
- The length of the message measures the amount of **surprise** on receiving the message
- Shannon's entropy is the expected length of the message to communicate a random variable  $X$

$$H_X = \mathbb{E}_X[-\log_2(\mathbb{P}(X))] = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))$$

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- The expected length is a measure of the uncertainty (how much information on average we need to convey the outcome)

# Real Codes

- Those of you with some computer science background will realise that we can't actually use different length strings in a code without paying some price
- We won't know where a code word ends so we can't decode the message
- An optimal solution is to use Huffman encoding where we associate the leaf of a tree with each code word
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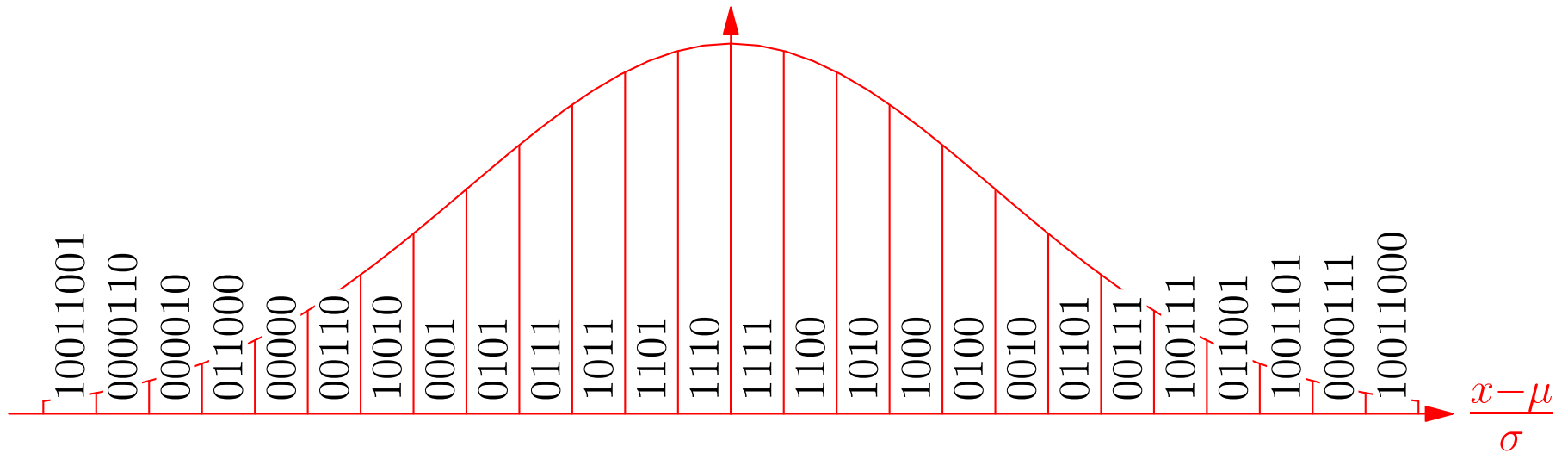
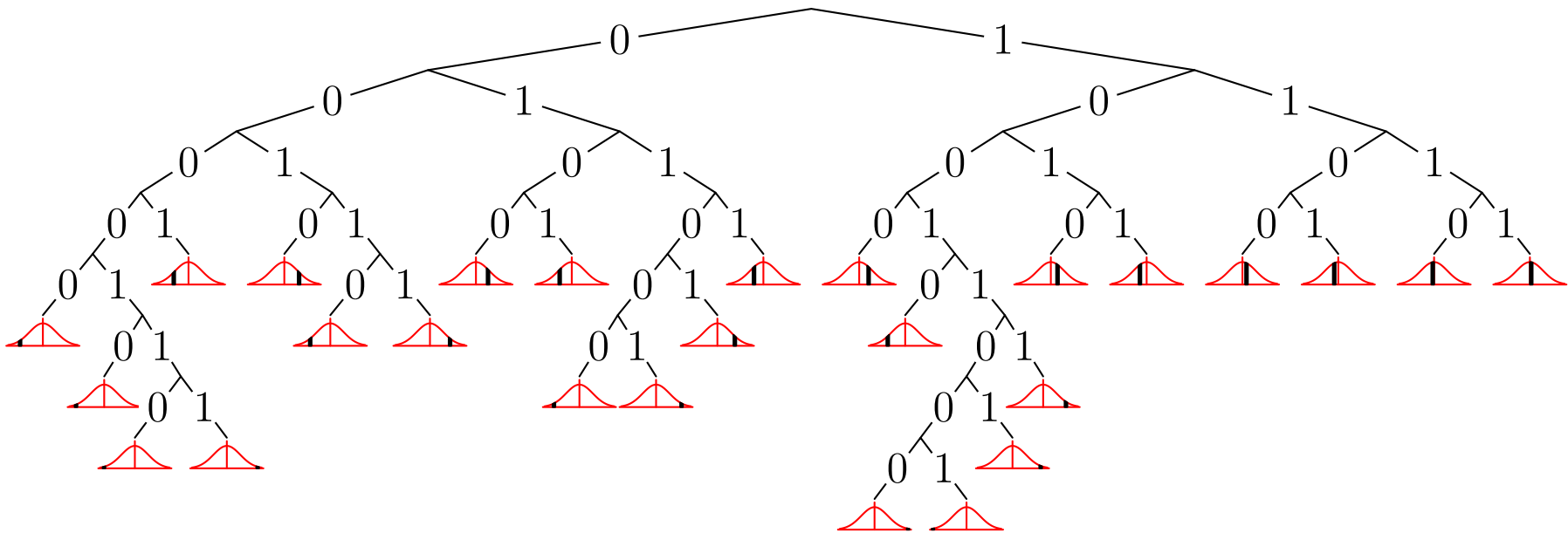
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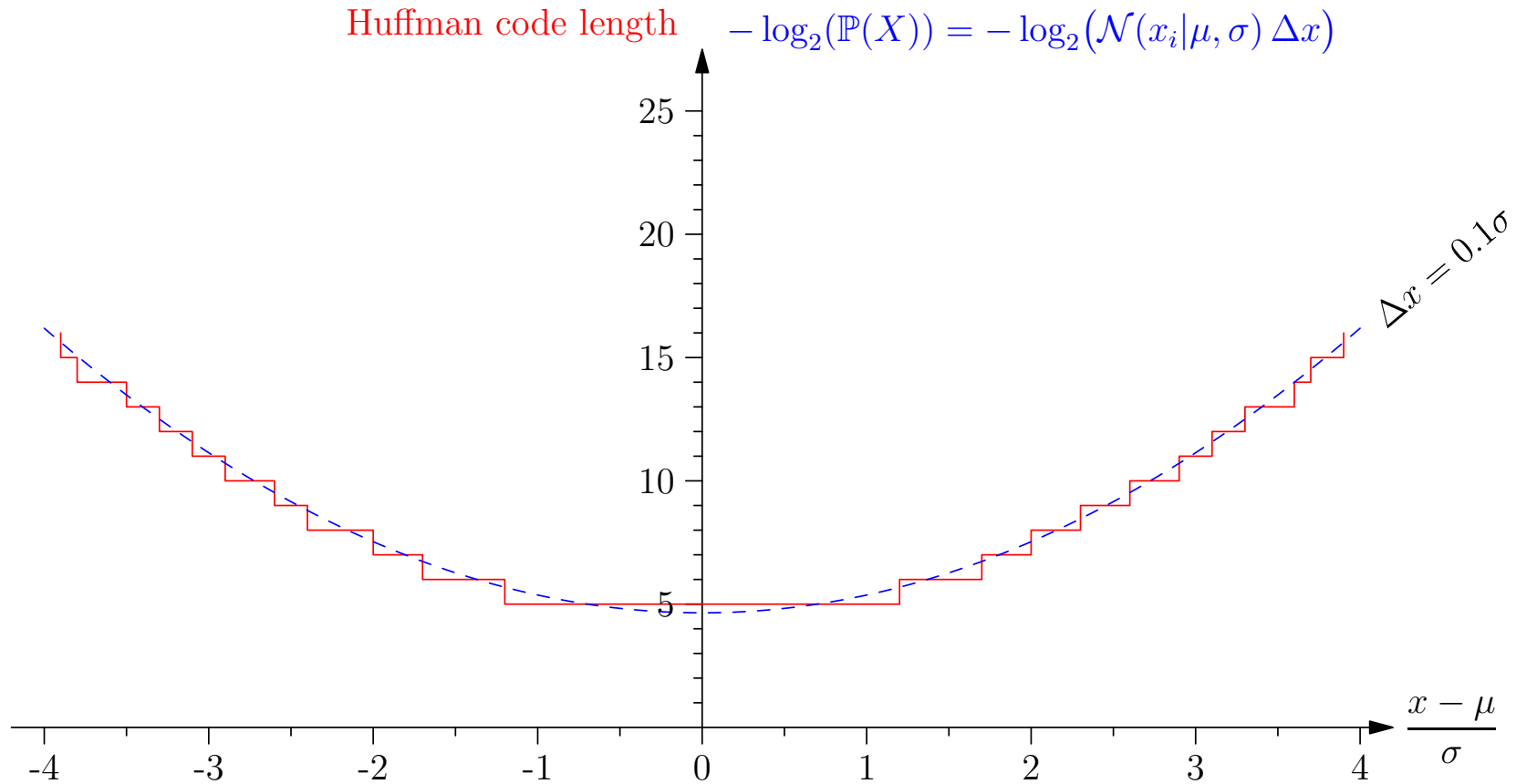
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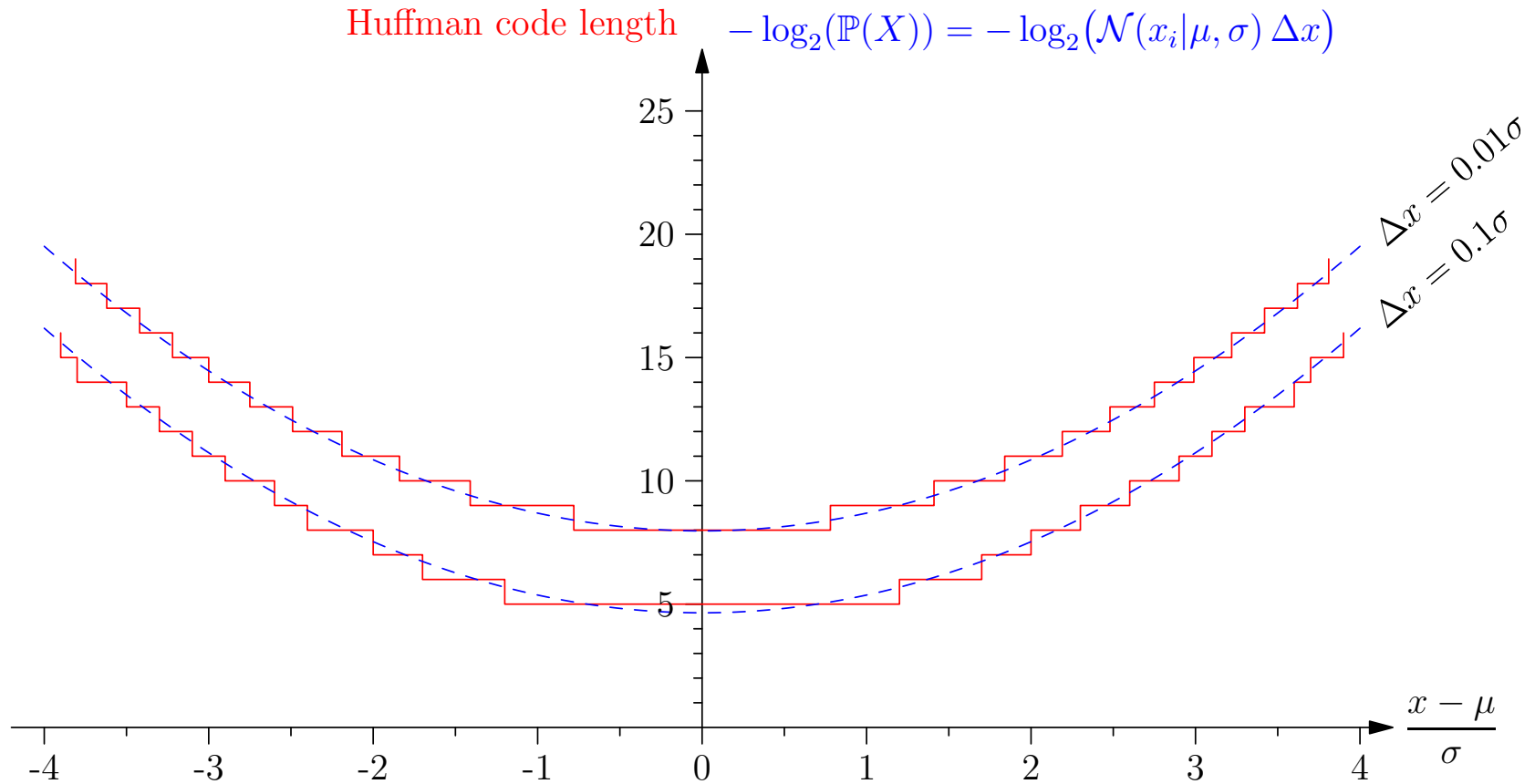
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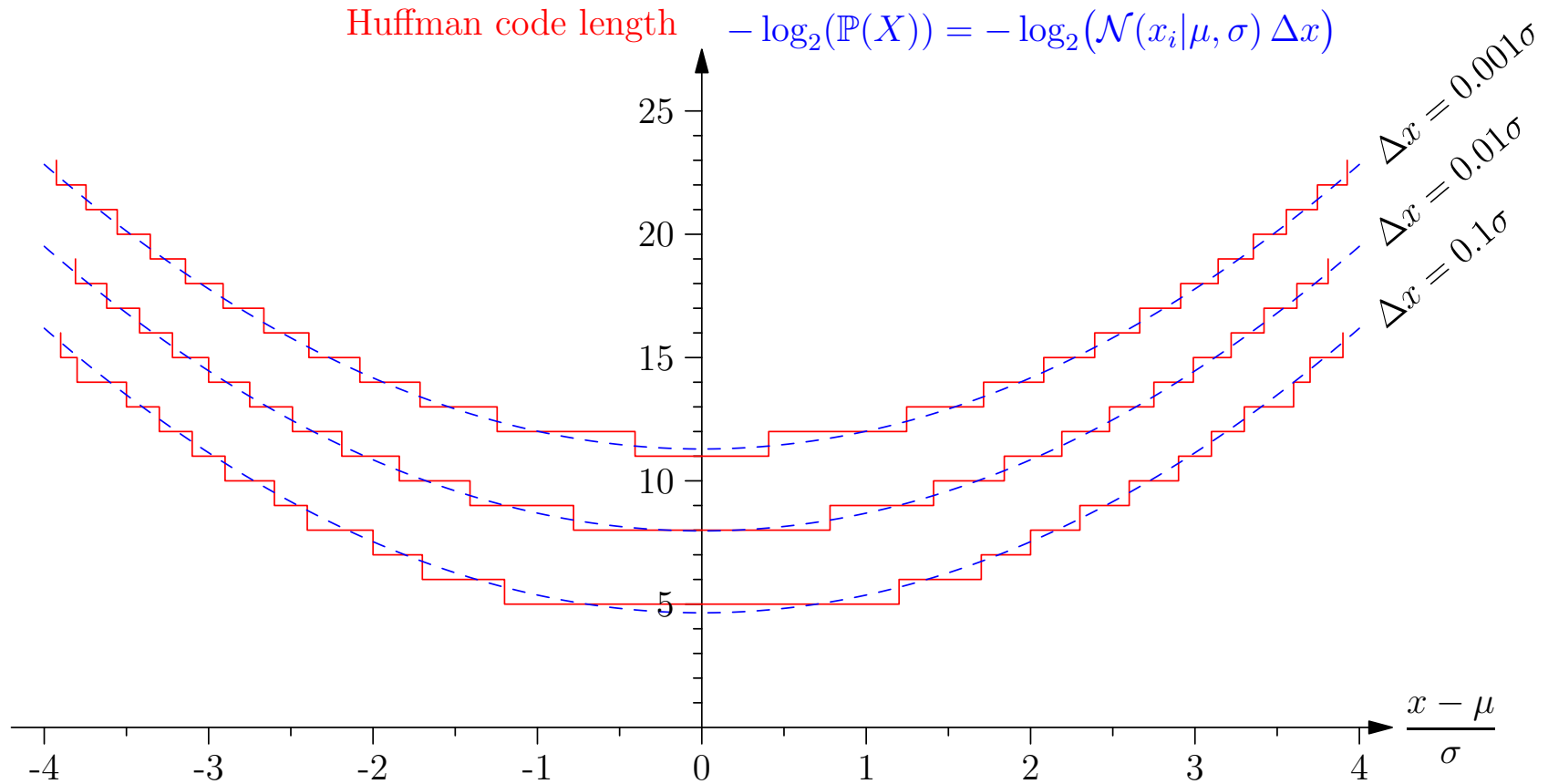
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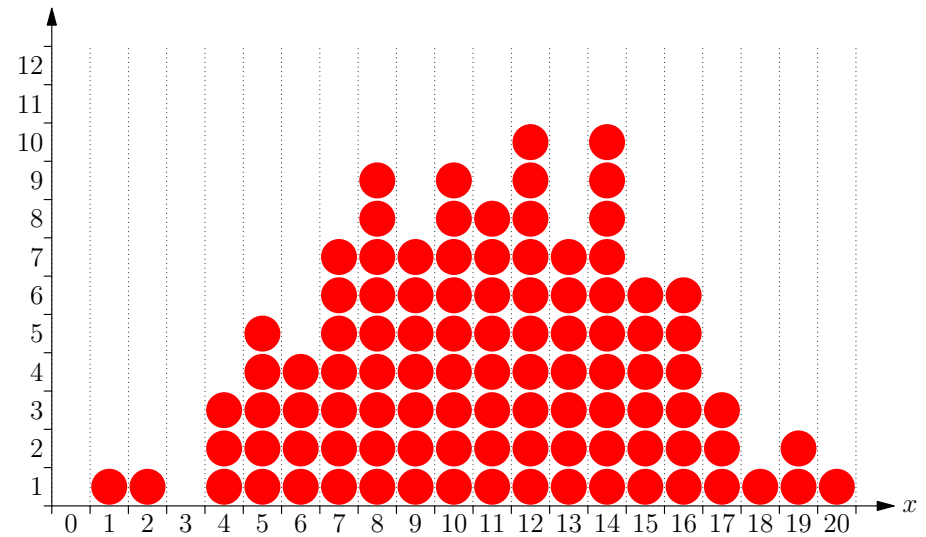
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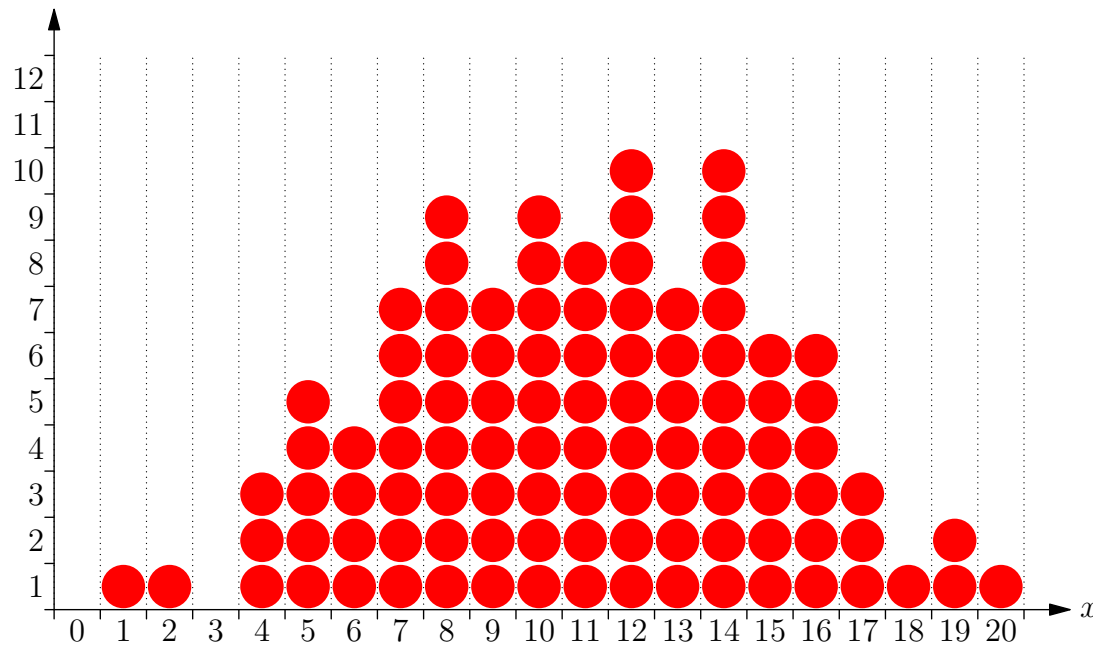
# Outline

1. Measuring Uncertainty
2. Code Length
3. **Maximum Entropy**



# Number of States

- Suppose I have  $N$  balls I then put in  $K$  boxes so that  $n_i$  balls finish up in box  $i$



- Counting the number of ways to do this

$$\mathbb{P}(\mathbf{n}) = \frac{N!}{n_1!n_2!\cdots n_K!}$$

# Stirling's Approximation

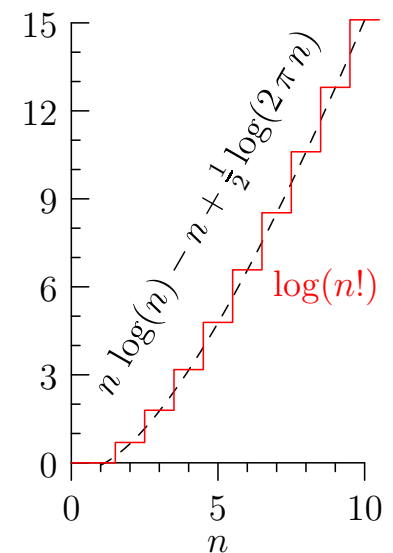
- We can approximate the factorial  $n!$  using **Stirling's approximation**

$$n! \approx \sqrt{2\pi n} n^n e^{-n}$$

$$\log(n!) = n \log(n) - n + \frac{1}{2} \log(2\pi n)$$

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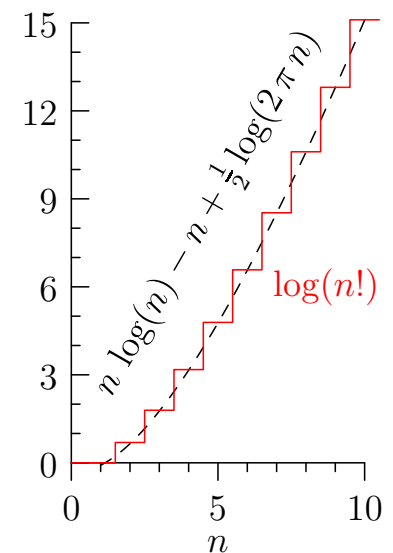
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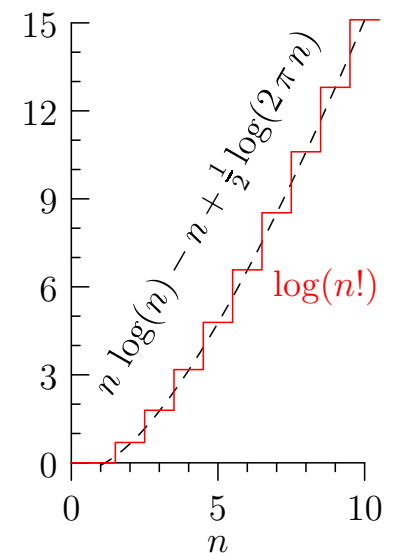
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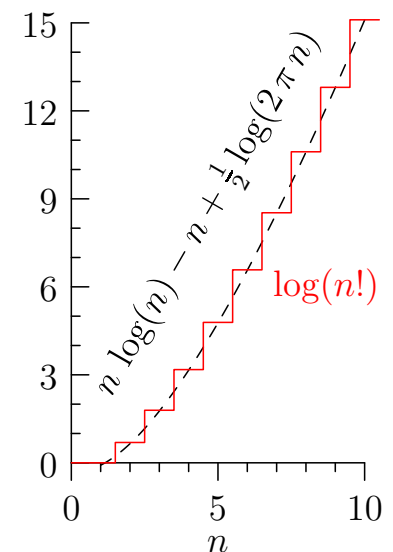
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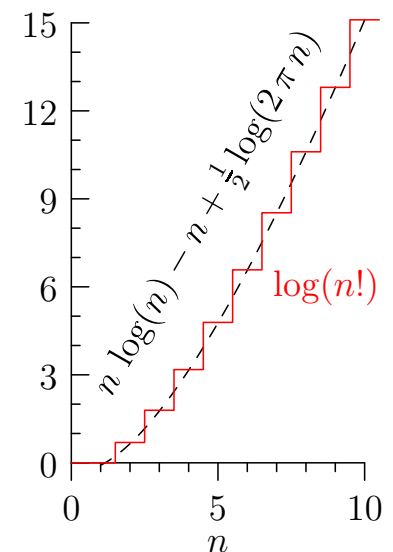
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# Number of States and Entropy

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- That is, the “entropy” can be seen as a measure of the logarithm of the number of configurations
- When the number of balls,  $N \rightarrow \infty$  the overwhelmingly likely configurations is the one that maximises the entropy subject to the observed mean and variance

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# Maximum Entropy Method

- When we are trying to infer a distribution given some observations then we can maximise the entropy subject to constraints
- This is known as the **maximum entropy method**
- We can rationalise this as this is by far the most likely set of configurations consistent with the observations
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# Knowing the Mean and Variance

- Consider a continuous random variable,  $X$ , with a known mean and second moment

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \mu_2 = \mu^2 + \sigma^2$$

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- Maximum entropy is often used to infer distributions
- It can be very effective, but it might not work well if there are other constraints that we have not included
- The place that they work superbly well is in statistical physics
- The whole of statistical physics is about inferring distributions making observations of volume, pressure, etc.
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# Conclusion

- Entropy provides a measure of the disorder or uncertainty in a system
- It forms the basis of information theory which we will look at in the next lecture
- $-\log(\mathbb{P}(X = x))$  can be seen as the minimum length of a message to communicate  $x$
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